NIOSH Science Blog

N95 Respirator Use During Pregnancy – Findings from Recent NIOSH Research

June 18, 2015 by Raymond Roberge, MD, MPH; Jung-Hyun Kim, PhD; and Jeffrey B. Powell, MS

Recent NIOSH research has shed some light on the topic of the safety of N95 filtering facepiece respirators (FFR) use by pregnant workers. Women make up approximately one-half of the US work force. At any given time, about 10% of those female workers of child-bearing age (15–44 years of age) will be pregnant. Because many women are employed in occupations that require the use of protective facemasks, such as medical/surgical masks and FFR, NIOSH conducted research into the safety of FFR use while pregnant. The most frequently used FFR in the US is the N95 FFR (commonly referred to as "N95 mask"), but little information was previously available about the safety of N95 FFR use during pregnancy. Some individuals complain of difficulty breathing when wearing an N95 FFR or other protective facemasks, and many pregnant women find that they become somewhat shorter of breath as their pregnancy progresses, causing concern that use of N95 FFRs during pregnancy might make breathing even more difficult and possibly harm the woman and her fetus. Beyond the issue of use by pregnant working women on the job, the question also has implications for pregnant women outside the workplace. People sometimes use N95 FFRs as a matter of personal choice during infectious disease outbreaks, during environmental disasters that pollute the air, and even in more common recreational activities that may expose them to airborne allergens, such as gardening and woodworking.

NIOSH Research – *Monitoring Oxygen Saturation and Transcutaneous Carbon Dioxide Levels of Pregnant Workers Who Wear Protective Facemasks*

In these studies[1,2] NIOSH researchers compared 22 healthy, non-smoking women who were in either their second or third trimesters of pregnancy with 22 healthy, non-smoking, non-pregnant women who were similar in age, height and weight. All of the women were experienced protective facemask users and were tested for one hour with and without wearing an N95 FFR while sitting, pedaling an exercise bicycle and standing.

Results

There were no significant differences between the effects of wearing an N95 FFR on pregnant and non-pregnant women with respect to their heart rate and function, breathing rate, percentage of oxygen and carbon dioxide in their arteries, ear temperature, and blood pressure, as well as their impressions of any warmth or exertion associated with the respirator. There was also no difference in the fetal heart rate with, and without, wearing a respirator.

The NIOSH study shows that the effects of wearing N95 FFRs are mild (average of one breath-per-minute decrease in the breathing rate, two beats-per-minute increase in heart rate, 1 – 7 mm Hg increase in diastolic blood pressure and 1- 2 mm increase in mean arterial pressure) and not different between pregnant and non-pregnant women. This is probably due to the fact that the filters of modern N95 FFR are able to be made thinner because they have electrically charged fibers that make them more efficient at trapping particles that are in the air, while at the same time being easier to breathe through. NIOSH researchers have also previously[3] shown that medical/surgical masks have roughly the same impact as N95 FFR when worn for one hour, therefore it is likely that they too would be safe for use by pregnant women.

There are other reasons why some individuals may find it harder to breathe when wearing an N95 FFR or medical/surgical mask, such as feelings of anxiety or claustrophobia, an uncomfortable warmth in the region of the face that is covered by the N95 FFR, and a change from normal nose breathing to mouth breathing that may occur. Also, those who have lung problems such as poorly-controlled asthma or chronic bronchitis may find it difficult to breathe when wearing an N95 FFR. These tests were only carried out for one hour and more studies are needed to find out if there are any additional effects from wearing an N95 FFR for longer periods of time. Any pregnant worker who is required to wear a protective facemask at work should first check with a qualified health professional to determine if any contraindications to wearing it exist.

Summary of Key Findings

- The effects of wearing an N95 FFR for one hour are similar for healthy pregnant and non-pregnant women.

- Wearing an N95 FFR for one hour by healthy pregnant women does not have an effect on the fetal heart rate.
- Similar effects would be expected with medical/surgical masks.

Raymond Roberge, MD, MPH; Jung-Hyun Kim, PhD; and Jeffrey B. Powell, MS

Dr. Roberge is a Research Medical Officer with the NIOSH National Personal Protective Technology Laboratory.

Dr. Kim is a Physical Scientist with the NIOSH National Personal Protective Technology Laboratory.

Mr. Powell is a Physical Scientist with the NIOSH National Personal Protective Technology Laboratory.

References:

1. Roberge RJ, Kim J-H, Powell JB. [2014] N95 respirator use during advanced pregnancy. American Journal of Infection Control 42: 1097-1100.
2. Kim J-H, Roberge RJ, Powell JB. [2015]. Effect of External Airflow Resistive Load on Postural and Exercise-associated Cardiovascular and Pulmonary Responses in Pregnancy. BMC Pregnancy and Childbirth 2015;15:45-52.
3. Roberge RJ, Kim J-H, Benson SM. [2012] Absence of consequential changes in physiological, thermal and subjective responses from wearing a surgical mask. Respiratory Physiology & Neurobiology 181: 29-35.

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Personal Protective Equipment, Reproductive Health, Respirators, Respiratory Health, Women

15 comments on “N95 Respirator Use During Pregnancy – Findings from Recent NIOSH Research”

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nice article .. Thank you .. !!

EXCELLENT! OUTSTANDING!! NIOSH’s 2013 guidelines on occupational lifting in pregnancy cleared up lots of outdated practices – and now this – WOW!!! This research will be a substantial aid in determining reasonable accommodations to protect pregnant workers and their unborn child (term used in OSHA 2012 HCS). With pregnant workers fairness laws sweeping across the country, timing for this research finding couldn’t be better. Can’t wait for NIOSH research to protect unborn child (again, OSHA term) from workplace noise. Thank you much, NIOSH.

thank you for information.

NIOSH research about pregnancy is very helpful. They show the real scenery. Thanks NIOSH for sharing this amazing guidelines.

If there is no real measurable medical change found while wearing the mask, why does OSHA have a huge questionnaire and certification standard? I am looking for something to substantiate the need for the amount of questions and depth of medical history required to be cleared before wearing a mask. Where is the evidence that concludes that being a diabetic or having a history of broken ribs or having an arrhythmia will cause distress enough to not allow a respirator to be used for a certain period of time? I would love some references or studies if anyone has that, I wasn’t able to find the answers.

> The purpose of this blog was to present new findings on respiratory use during pregnancy. The questionnaire that is Appendix C of the OSHA Standard 1910.134, was promulgated by OSHA in 1998. The rationale for including the medical history questionnaire can be found in the Preamble of the January 1998 Final Rule which can be found in the “VII. Summary and Explanation” section at: https://www.osha.gov/pls/oshaweb/owasrch.search_form?p_doc_type=PREAMBLES&p_toc_level=1&p_keyvalue=Respiratory~Protection. Many exhibits are referenced in this section which may be available from OSHA.

Why is it a yearly required testing. In Canada it is a every 2 year requirement.

The article about the use of N95 Respirator during pregnancy.This is new knowledge for me.

Please see the study

https://pubmed.ncbi.nlm.nih.gov/26579222/

Conclusions: Breathing through N95 mask materials have been shown to impede gaseous exchange and impose an additional workload on the metabolic system of pregnant healthcare workers, and this needs to be taken into consideration in guidelines for respirator use. The benefits of using N95 mask to prevent serious emerging infectious diseases should be weighed against potential respiratory consequences associated with extended N95 respirator usage.

> Thank you! You read my mind. I actually found and read the study before I found this synopsis by the CDC and wondered why they left out the rest of the information for the sake of painting a picture that there's zero side effects or differences. Very misleading and everyone should do their own diligence and read the actual study, not someone's interpretation of it — clearly.

> OSHA requires that an employee undergo a medical evaluation to verify that it is safe for them to use a respirator. We encourage discussion about medical concerns or conditions (including pregnancy) with your physician.

Let's be honest, they didn't do the study for a 12 hour shift when we're half suffocating from heat and lack of air. Lol

> An N95 doesn't lower SpO2 while pregnant women use an excercise bike for an hour, but surgical masks worn by male surgeons might. Interesting.... 😳
> https://www.researchgate.net/publication/28219427_Preliminary_report_on_surgical_mask_induced_deoxygenation_during_major_surgery

> It's easy to wear it for 1 hour but how about 12!!!!!! Or even 8! Shame on them that they don't do that kind of studies and protect women!

> Our study did not assess the effects of wearing an N95 on pregnant healthcare workers at higher work intensities and over longer durations. This is because such a study would not meet the ethical requirements for research involving pregnant women or fetuses which are outlined in Federal Regulations 45 CFR § 46.204 – "Research involving pregnant women or fetuses."

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